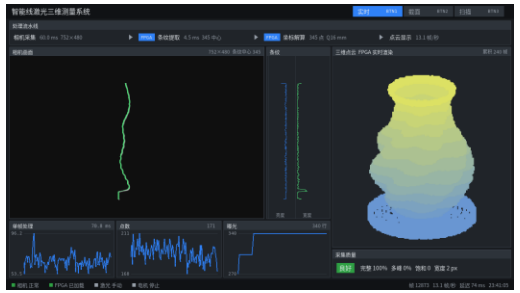
智能线激光
三维测量系统

作品简介

面向小型零件的非接触轮廓与尺寸检测。单块PYNQ-Z2完成条纹提取、三维重建与HDMI点云显示。



BTN1 实时

激光中心
定点三维坐标
实时点云



BTN2 截面

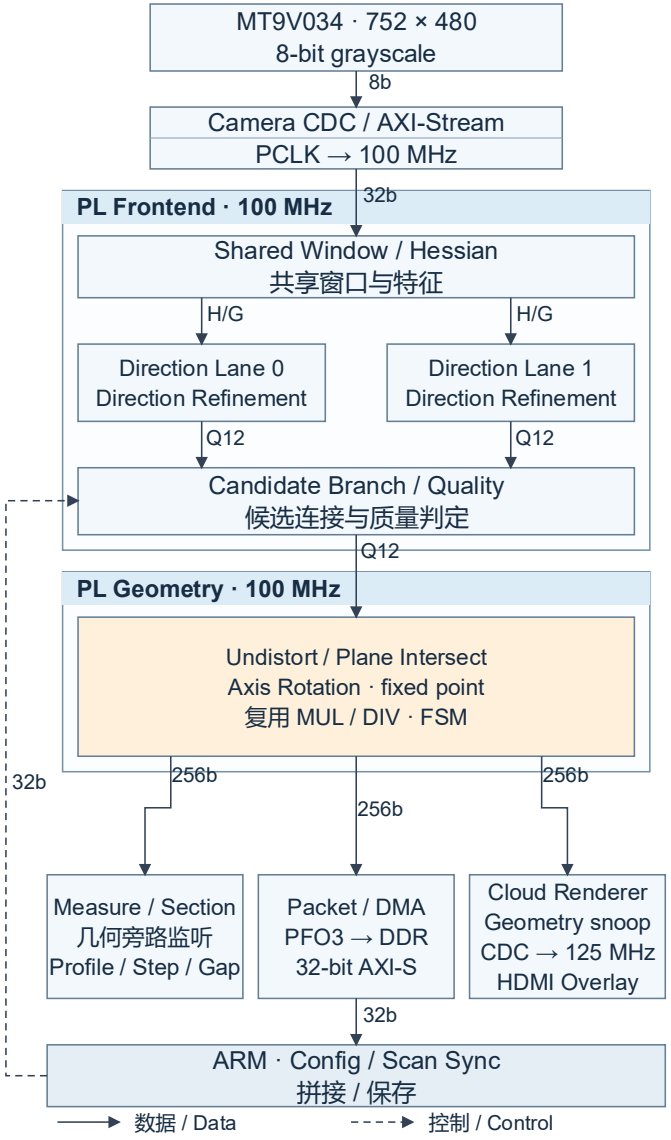
台阶高度
间隙提示
噪声与趋势



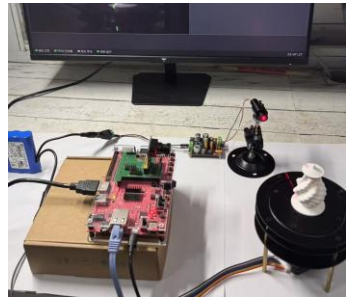
BTN3 扫描

角度进度
点云累积
结果导出

PYNQ-Z2 单板运行
752 × 480 灰度输入



创新设计



板上实测：PYNQ-Z2

测试结果

PL图像处理能力

26.5 帧 / 秒

逐帧处理，无错误包

条纹中心误差中位数 **0.02 像素**

已知中心位置的合成条纹

一圈扫描：200视角 / 约2.5分钟 / 约6.6万个点

测量项目	实测结果
台阶标准值	2 / 3 / 5 mm
台阶实测值	**2.00 / 2.90 / 4.96 mm**
阶梯圆柱直径偏差	−0.32 to −0.10 mm
花瓶颈部直径偏差	−0.27 to +0.05 mm

直径偏差为扫描值减卡尺读数。
阶梯圆柱结果用于标定复核。

FPGA实现资源

LUT 73.5%　FF 47.8%
BRAM 78.9%　DSP 60.9%
WNS +0.230 ns　片上功耗约2.37 W

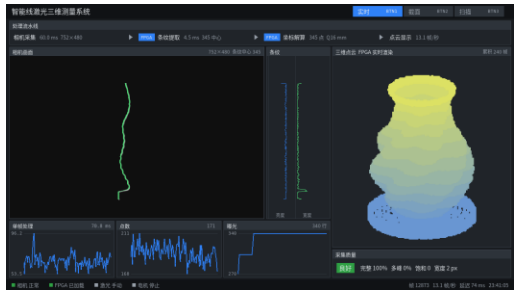


阶梯圆柱实测点云　花瓶点云重建表面

Intelligent Line-Laser 3D Measurement System

INTRODUCTION

Non-contact profile and dimension inspection of small parts.
Pixel extraction, 3D geometry and HDMI point-cloud rendering run on one PYNQ-Z2.



BTN1 LIVE

Laser centres
Fixed-point XYZ
Live cloud



BTN2 SECTION

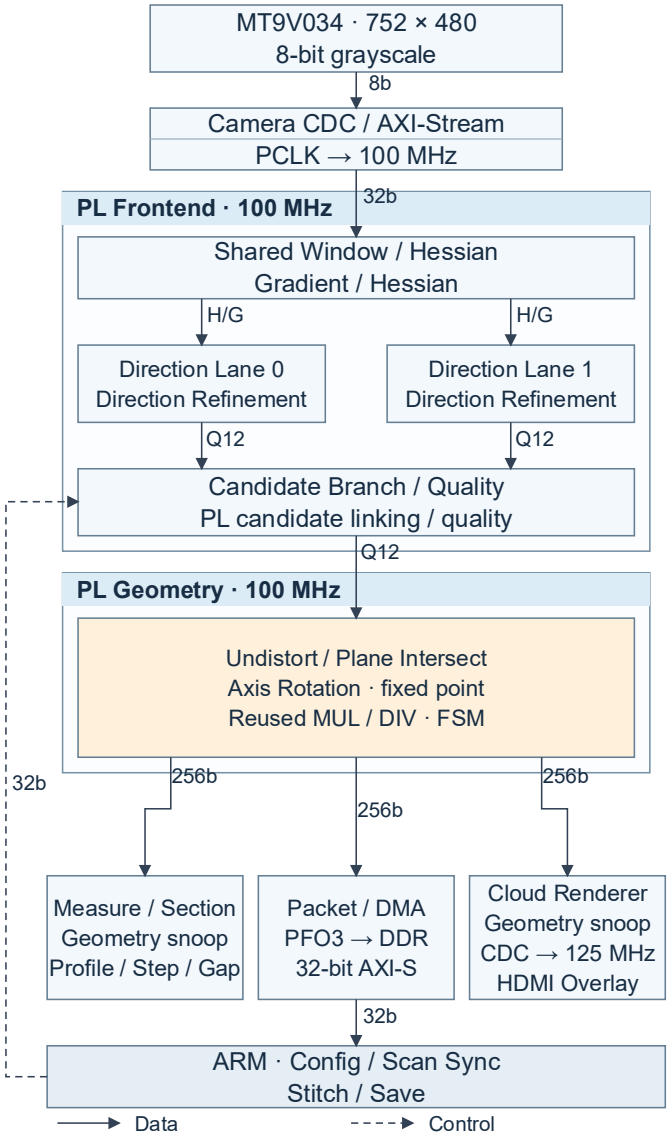
Step heights
Gap indication
Noise and trend



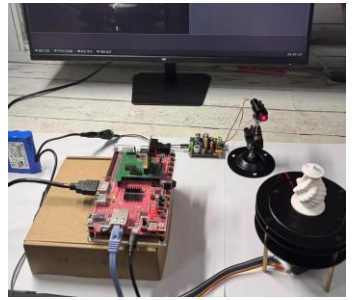
BTN3 SCAN

Angle progress
Cloud accumulation
Result export

Single-board operation
752 × 480 grayscale input



CREATIVE DESIGN



On-board test: PYNQ-Z2

RESULTS

PL image processing

26.5 frames/s

Frame-by-frame processing, zero packet errors

Median stripe-centre error **0.02 px**

Synthetic stripes with known centre positions
360° scan: 200 views / ≈2.5 min / ≈66k points

Measurement	Result
Steps, nominal	2 / 3 / 5 mm
Steps, measured	**2.00 / 2.90 / 4.96 mm**
Cylinder diameter error	−0.32 to −0.10 mm
Vase neck error	−0.27 to +0.05 mm

Signed diameter errors against caliper readings.
Cylinder results are calibration rechecks.

Implemented FPGA resources

LUT 73.5% FF 47.8%
BRAM 78.9% DSP 60.9%
WNS +0.230 ns On-chip power ≈2.37 W



Cylinder scan cloud Vase scan mesh